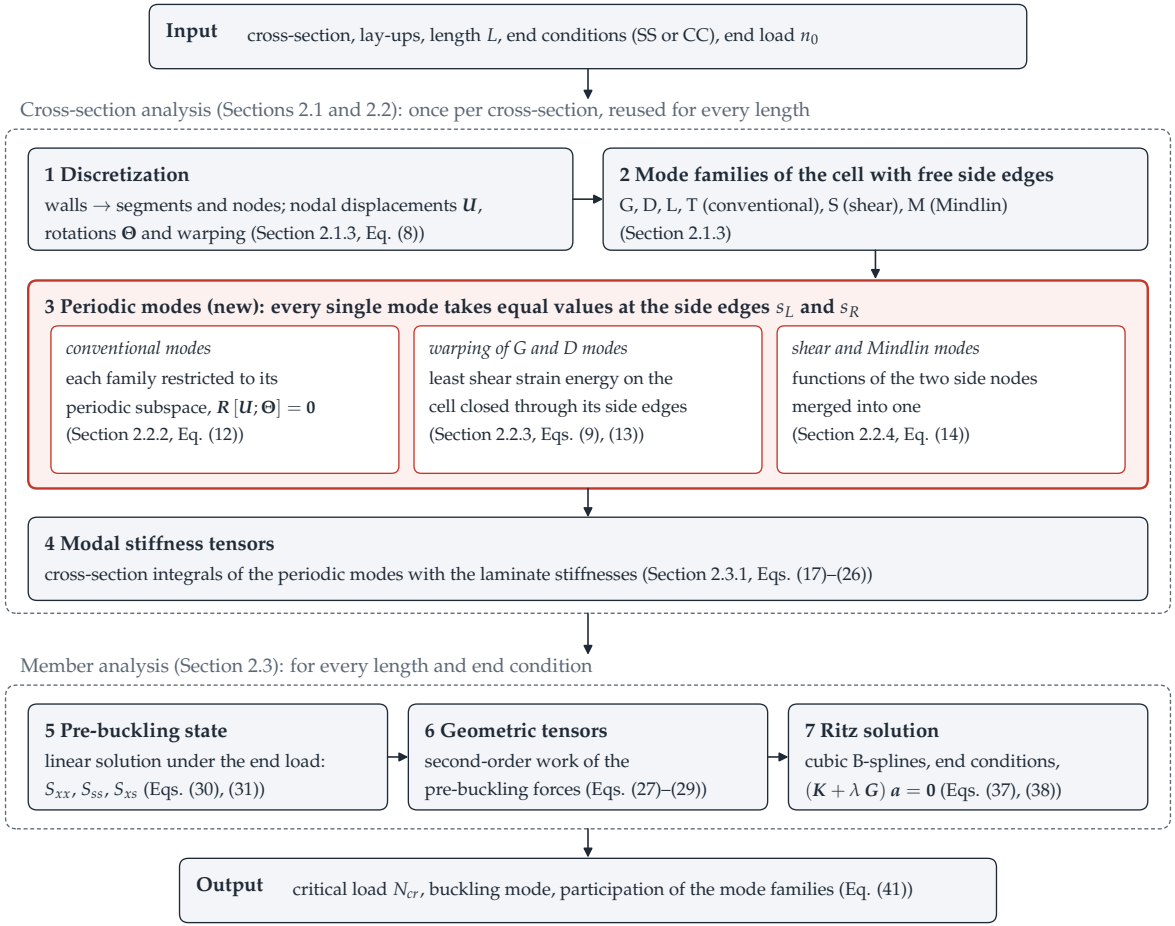

Input
cross-section, lay-ups, length L, end conditions (SS or CC), end load n_0
Cross-section analysis (Sections 2.1 and 2.2): once per cross-section, reused for every length
1 Discretization
walls → segments and nodes; nodal displacements $\boldsymbol{U}$, rotations $\boldsymbol{\Theta}$ and warping (Section 2.1.3, Eq. (8))
2 Mode families of the cell with free side edges
G, D, L, T (conventional), S (shear), M (Mindlin) (Section 2.1.3)
3 Periodic modes (new): every single mode takes equal values at the side edges s_L and s_R
conventional modes
each family restricted to its periodic subspace, $\boldsymbol{R}[\boldsymbol{U};\boldsymbol{\Theta}] = \boldsymbol{0}$ (Section 2.2.2, Eq. (12))
warping of G and D modes
least shear strain energy on the cell closed through its side edges (Section 2.2.3, Eqs. (9), (13))
shear and Mindlin modes
functions of the two side nodes merged into one (Section 2.2.4, Eq. (14))
4 Modal stiffness tensors
cross-section integrals of the periodic modes with the laminate stiffnesses (Section 2.3.1, Eqs. (17)–(26))
Member analysis (Section 2.3): for every length and end condition
5 Pre-buckling state
linear solution under the end load: S_{xx}, S_{ss}, S_{xs} (Eqs. (30), (31))
6 Geometric tensors
second-order work of the pre-buckling forces (Eqs. (27)–(29))
7 Ritz solution
cubic B-splines, end conditions, $(\boldsymbol{K} + \lambda\, \boldsymbol{G})\, \boldsymbol{a} = \boldsymbol{0}$ (Eqs. (37), (38))
Output
critical load N_{cr}, buckling mode, participation of the mode families (Eq. (41))